STAT3888: Full Population Analysis for Q1B

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Going back to the following problem:

> **Research Problem**
>
> Does meeting physical activity guidelines compared to those who do not, improve liver biomarkers in smokers and medium/high risk alcohol consumers, after accounting for age, sex, waist circumference, and number of daily smokers in the household?

We would like to see if including the entire adult population (not just smokers or medium/high risk alcohol consumers) can provide us with better analysis. This document will likely be shorter as a lot of the analysis has been done in `FeasibilityAnalysisQ1b` and I don't want to just replicate everything there, as it is quite extensive.

Full Population Data Set

We will just call the data sets again as usual, and then not filter out the sub-population to get more subject entries after the merge, and also I whill use `WAIST` as the waist circumference, which will combine the female and male waist circumference variables, to reduce the number of `NA` values and redundancy as well.

▼ Code

```
# Original data sets
bsp = readRDS(here("Objects", "AHSnhs11bsp.rds"))
bhh = readRDS(here("Objects", "AHSnhs11bhh.rds"))
bbi = readRDS(here("Objects", "AHSnhs11bbi.rds"))

# Variable lists
bbi.vars = c("ABSLID", "ABSPID", "ALTRESB", "GGTRESB")
bsp.vars = c("ABSLID", "ABSPID", "AL2K7DAY", "SMKAGEDB", "EXREGU
             "PHDWCATM", "PHDWCATF")
bhh.vars = c("ABSLID", "DSMKHHRB")

# Taking subsets of variables and relevel
bbir = bbi %>% select(all_of(bbi.vars)) %>%
  mutate(ALTRESB = fct_relevel(ALTRESB, levels(bbi$ALTRESB)[9]))
```

```
bspr = bsp %>% select(all_of(bsp.vars)) %>%
  mutate(EXREGUIC = fct_relevel(EXREGUIC, levels(bsp$EXREGUIC)[2
         AGEB = fct_relevel(AGEB, levels(bsp$AGEB)[11], after =
         PHDWCATM = fct_relevel(PHDWCATM, "Not at risk", "Increa
                                "Substantially increased risk"),
         PHDWCATF = fct_relevel(PHDWCATF, "Not at risk", "Increa
                                "Substantially increased risk"),
         WAIST = ifelse(!is.na(PHDWCATM),
                        as.character(PHDWCATM), as.characte
         WAIST = factor(WAIST)) %>%
  select(-c(PHDWCATM, PHDWCATF))
bhhr = bhh %>% select(all_of(bhh.vars)) %>%
  mutate(DSMKHHRB = fct_relevel(DSMKHHRB, "None", "One", "Two",

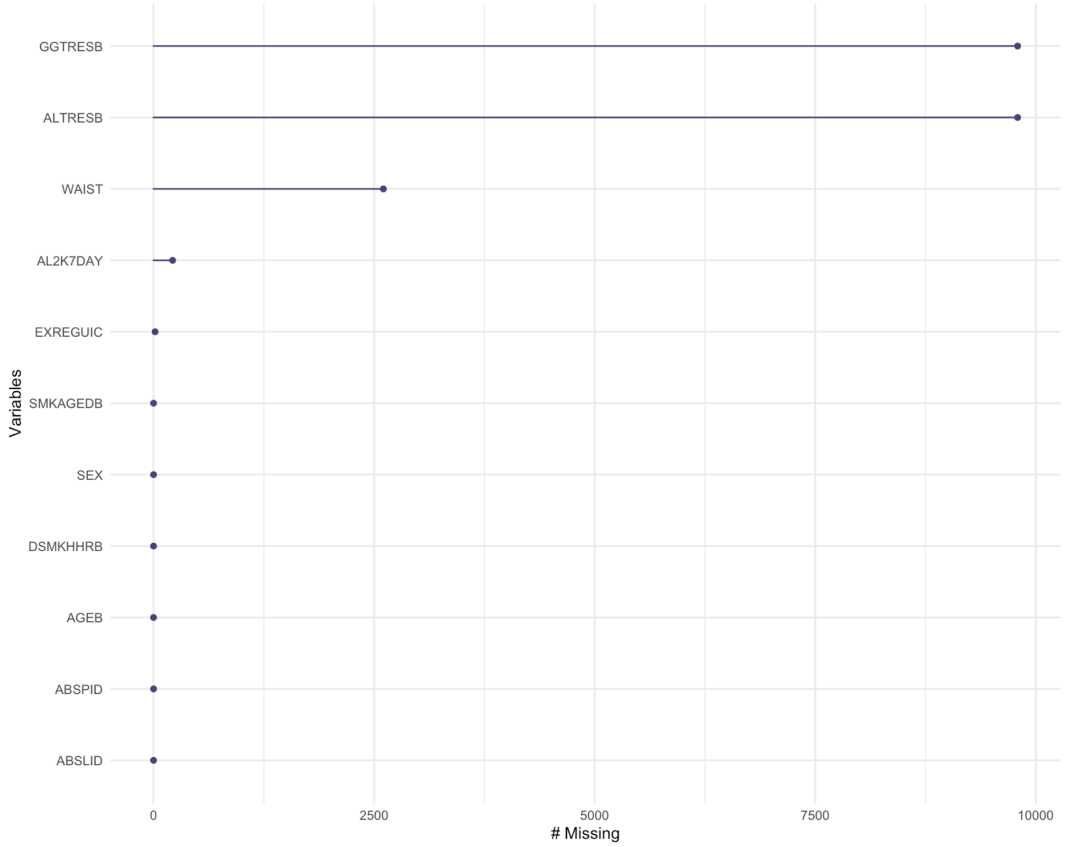
# Population data (no filtering this time)
fdat = inner_join(bspr, bbir, by = c("ABSLID", "ABSPID")) %>% d
fdat.full = inner_join(fdat, bhhr, by = "ABSLID") %>%
  filter(!AGEB %in% c("0-4 years", "5-9 years", "10-14 years", "
  mutate(AGEB = fct_drop(AGEB))

noquote(paste("Dimensions of FDAT.FULL:", dim(fdat.full)[1], "x"
```

```
[1] Dimensions of FDAT.FULL: 15475 x 11
```

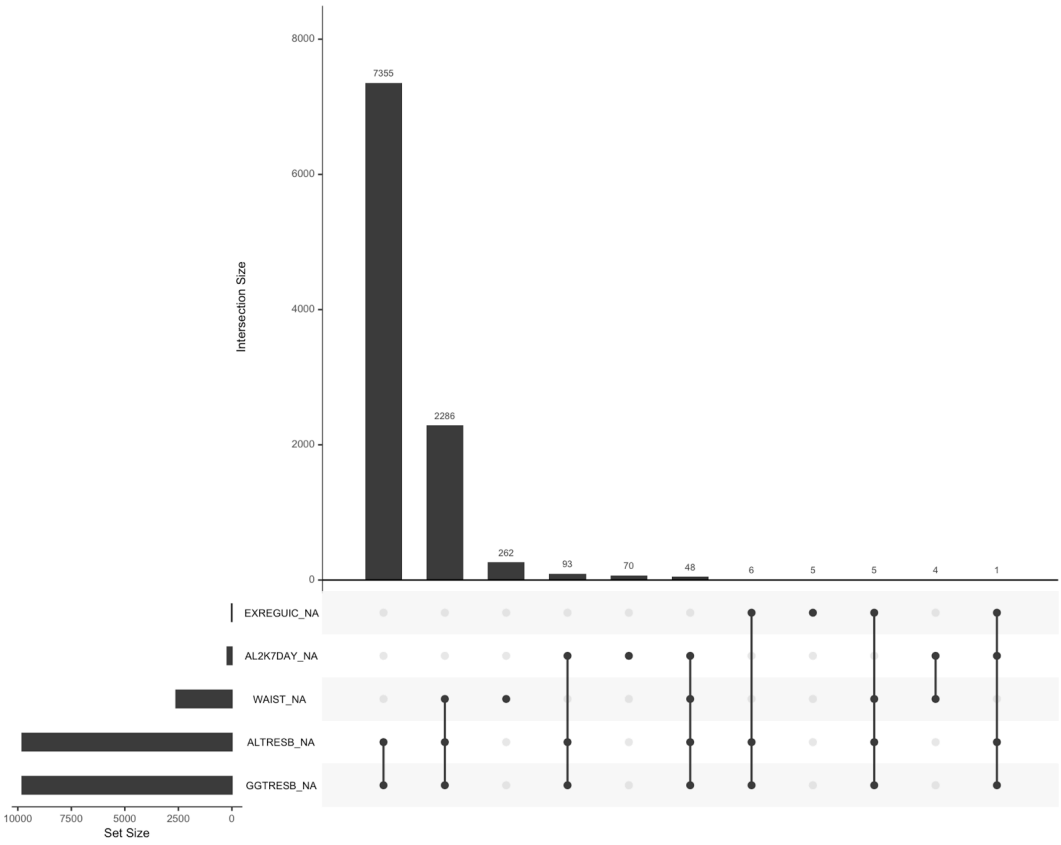
▼ Code

```
gg_miss_var(fdat.full)
```



▼ Code

```
gg_miss_upset(fdat.full)
```



▼ Code

```
miss_var_summary(fdat.full)
```

```
# A tibble: 11 × 3
   variable n_miss pct_miss
   <chr>     <int>    <num>
 1 ALTRESB    9794   63.3
 2 GGTRESB    9794   63.3
 3 WAIST      2605   16.8
 4 AL2K7DAY    216    1.40
 5 EXREGUIC     17    0.110
 6 ABSLID        0    0
 7 ABSPID        0    0
 8 SMKAGEDB      0    0
 9 AGEB          0    0
10 SEX           0    0
11 DSMKHHRB      0    0
```

χ^2 Tests for Independence

▼ Code

```
# Test 1; check expected counts, and print output
test1 = chisq.test(table(fdat.full$EXREGUIC, fdat.full$ALTRESB)
noquote(paste("Test 1 - Any expected counts < 5:", any(test1$exp
```

```
[1] Test 1 - Any expected counts < 5: FALSE
```

▼ Code

```
test1
```

```

    Pearson's Chi-squared test

data:  table(fdat.full$EXREGUIC, fdat.full$ALTRESB)
X-squared = 17.934, df = 9, p-value = 0.03595
```

▼ Code

```
# Test 2
test2 = chisq.test(table(fdat.full$EXREGUIC, fdat.full$GGTRESB)
noquote(paste("Test 2 - Any expected counts < 5:", any(test2$exp
```

```
[1] Test 2 - Any expected counts < 5: FALSE
```

▼ Code

```
test2
```

```

    Pearson's Chi-squared test

data:  table(fdat.full$EXREGUIC, fdat.full$GGTRESB)
X-squared = 72.297, df = 11, p-value = 4.46e-11
```

We see in the case of the full population, the significance flips; there seems to be statistically significant a relationship between physical activity and both liver biomarkers.

Odds Ratio and Relative Risk

We will have to use the truncated categories for ALT and GGT for this one; trying to do it with *all* levels of `ALTRESB` and `GGTRESB` causes mathematical instability,

and at best we can use the three-category versions.

▼ Code

```
fdat.full = fdat.full %>%
  mutate(
    ALT3 = case_when(
      SEX == "Male" & ALTRESB %in% c("Less than 10",
                                     "10 to less than 15",
                                     "15 to less than 20",
                                     "20 to less than 25",
                                     "25 to less than 30") ~ "Lo
      SEX == "Male" & ALTRESB %in% c("30 to less than 35",
                                     "35 to less than 40") ~ "Bo
      SEX == "Male" & ALTRESB %in% c("40 to less than 45",
                                     "45 to less than 50",
                                     "More than 50") ~ "High",

      SEX == "Female" & ALTRESB %in% c("Less than 10",
                                       "10 to less than 15",
                                       "15 to less than 20",
                                       "20 to less than 25") ~ "Lo
      SEX == "Female" & ALTRESB == "25 to less than 30" ~ "Borde
      SEX == "Female" & ALTRESB %in% c("30 to less than 35",
                                       "35 to less than 40",
                                       "40 to less than 45",
                                       "45 to less than 50",
                                       "More than 50") ~ "High",
      TRUE ~ NA
    ) %>% as.factor(),
    GGT3 = case_when(
      SEX == "Male" & GGTRESB %in% c("10 or less", "More than 10
                                     "More than 15 to 20",
                                     "More than 20 to 25",
                                     "More than 25 to 30",
                                     "More than 30 to 35",
                                     "More than 35 to 40",
                                     "More than 40 to 45") ~ "Lo
      SEX == "Male" & GGTRESB %in% c("More than 45 to 50",
                                     "More than 50 to 55") ~ "Bo
      SEX == "Male" & GGTRESB %in% c("More than 55 to 60",
                                     "More than 60") ~ "High",

      SEX == "Female" & GGTRESB %in% c("10 or less", "More than
                                       "More than 15 to 20",
                                       "More than 20 to 25",
                                       "More than 25 to 30") ~ "Lo
      SEX == "Female" & GGTRESB == "More than 30 to 35" ~ "Borde
```

```
      SEX == "Female" & GGTRESB %in% c("More than 35 to 40",
                                       "More than 40 to 45",
                                       "More than 45 to 50",
                                       "More than 50 to 55",
                                       "More than 55 to 60",
                                       "More than 60") ~ "High",
      TRUE ~ NA
    ) %>% as.factor(),
    EXREGUIC = relevel(EXREGUIC, ref = "Met recommended guideli
  ) %>%
  mutate(ALT3 = fct_relevel(ALT3, "Low", "Borderline", "High"),
         GGT3 = fct_relevel(GGT3, "Low", "Borderline", "High"))

# Odds Ratio and Relative Risk
mymat = table(fdat.full$EXREGUIC, fdat.full$ALT3)

# Run OR and RR Calculations
or_model = oddsratio(mymat)
rr_model = riskratio(mymat)

# Clean and extract only the relevant metrics
or_stats = or_model$measure[2, ]
rr_stats = rr_model$measure[2, ]

# Data frame for output
OR.RR = data.frame(
  Metric = c("Odds Ratio (OR)", "Relative Risk (RR)"),
  Estimate = c(or_stats["estimate"], rr_stats["estimate"]),
  Lower_CI = c(or_stats["lower"], rr_stats["lower"]),
  Upper_CI = c(or_stats["upper"], rr_stats["upper"])
)

OR.RR %>%
  mutate(across(where(is.numeric), ~ round(., 2))) %>%
  kable(
    caption = "Odds Ratio and Relative Risk of EXREGUIC and ALT
    col.names = c("Measure", "Point Estimate", "Lower 95% CI", "
    align = "lccc"
  )
```

Odds Ratio and Relative Risk of EXREGUIC and ALT Levels

Measure	Point Estimate	Lower 95% CI	Upper 95% CI
Odds Ratio (OR)	0.96	0.81	1.13
Relative Risk (RR)	0.96	0.83	1.12

▼ Code

```
# For GGT as well
# Odds Ratio and Relative Risk
mymat = table(fdat.full$EXREGUIC, fdat.full$GGT3)

# Run OR and RR Calculations
or_model = oddsratio(mymat)
rr_model = riskratio(mymat)

# Clean and extract only the relevant metrics
or_stats = or_model$measure[2, ]
rr_stats = rr_model$measure[2, ]

# Data frame for output
OR.RR = data.frame(
  Metric = c("Odds Ratio (OR)", "Relative Risk (RR)"),
  Estimate = c(or_stats["estimate"], rr_stats["estimate"]),
  Lower_CI = c(or_stats["lower"], rr_stats["lower"]),
  Upper_CI = c(or_stats["upper"], rr_stats["upper"])
)

OR.RR %>%
  mutate(across(where(is.numeric), ~ round(., 2))) %>%
  kable(
    caption = "Odds Ratio and Relative Risk of EXREGUIC and GGT
    col.names = c("Measure", "Point Estimate", "Lower 95% CI", "
    align = "lccc"
  )
```

Odds Ratio and Relative Risk of EXREGUIC and GGT Levels

Measure	Point Estimate	Lower 95% CI	Upper 95% CI
Odds Ratio (OR)	1.39	1.11	1.75
Relative Risk (RR)	1.36	1.10	1.69

Controlling for Confounders

This uses the same CMH test as before.

▼ Code

```
# CMH Test Summary for ALT
```

```r
ordered_confounders = c("DSMKHHRB", "WAIST", "AGEB", "SEX")
cmh_results = list()

# Loop over confounders
for (i in seq_along(ordered_confounders)) {
  var_name = ordered_confounders[i]

  cat("\n====================================\n")
  cat("CMH Test for:", var_name, "\n")
  cat("====================================\n")

  # Generate contingency table over all three variables
  cmh_table = table(fdat.full$EXREGUIC, fdat.full$ALT3, fdat.fu

  # Run the test for each confounder
  test = tryCatch({
    mantelhaen.test(cmh_table)
  },
    error = function(e) {NULL}
  )

  print(test)
}
```

```
====================================
CMH Test for: DSMKHHRB
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 11.017, df = 2, p-value =
0.004053


====================================
CMH Test for: WAIST
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 5.1166, df = 2, p-value = 0.07744


====================================
```

```
CMH Test for: AGEB
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 15.82, df = 2, p-value = 0.000367


====================================
CMH Test for: SEX
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 12.642, df = 2, p-value =
0.001798
```

▼ Code

```r
# CMH Test Summary for GGT
ordered_confounders = c("DSMKHHRB", "WAIST", "AGEB", "SEX")
cmh_results = list()

# Loop over confounders
for (i in seq_along(ordered_confounders)) {
  var_name = ordered_confounders[i]

  cat("\n====================================\n")
  cat("CMH Test for:", var_name, "\n")
  cat("====================================\n")

  # Generate contingency table over all three variables
  cmh_table = table(fdat.full$EXREGUIC, fdat.full$GGT3, fdat.fu

  # Run the test for each confounder
  test = tryCatch({
    mantelhaen.test(cmh_table)
  },
    error = function(e) {NULL}
  )

  print(test)
}
```

```
====================================
CMH Test for: DSMKHHRB
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 44.557, df = 2, p-value = 2.111e-
10


====================================
CMH Test for: WAIST
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 23.934, df = 2, p-value = 6.351e-
06


====================================
CMH Test for: AGEB
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 43.291, df = 2, p-value = 3.977e-
10


====================================
CMH Test for: SEX
====================================

    Cochran-Mantel-Haenszel test

data:  cmh_table
Cochran-Mantel-Haenszel M^2 = 50.274, df = 2, p-value = 1.211e-
11
```

Some of the significances change around from the sub-population, so this could be something we can look into later on.

Logistic Regression

This will be the last part for now, I don't think I will do the imputation analysis for the full data set. We still need to use the multinomial logistic regression in this case, not due to mathematical instability but because we are using more than three categories. As mentioned in the other file, we use `nnet`'s implementation, which while it uses the MLE instead of Firth's penalised regression, it is much faster than `brglm2`'s implementation, and since we are using it in the other data set, we should keep using it for consistency.

▼ Code

```
# ALT
model_file = here("Objects", "Full_data_MLR_ALT3.rds")
if (file.exists(model_file)) {
  full.mlr.alt = readRDS(model_file)
} else {
  full.mlr.alt = multinom(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST +
                          data = fdat.full, Hess = TRUE)
  saveRDS(full.mlr.alt, model_file)
}
summary(full.mlr.alt)
```

```
Call:
multinom(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB,
    data = fdat.full, Hess = TRUE)

Coefficients:
           (Intercept) EXREGUICDid not meet recommended
guidelines
Borderline   -2.839259
-0.03914779
High         -2.476492
0.21277380
           AGEB20–24 years AGEB25–29 years AGEB30–34 years
AGEB35–39 years
Borderline      0.05564075       0.6990091      0.45928982
0.1255106
High            0.01969485       0.1589173     -0.03440286
0.3183683
           AGEB40–44 years AGEB45–49 years AGEB50–54 years
AGEB55–59 years
Borderline      0.28705795       0.7016212       0.7659134
0.39393957
```

```
High            0.04777947       0.2808256       0.2716942
-0.00408018
           AGEB60-64 years AGEB65-69 years AGEB70-74 years
AGEB75-79 years
Borderline      0.65365479       0.1219236      0.08788967
-1.151714
High           -0.09894756      -0.3519374     -0.92372002
-1.528199
           AGEB80-84 years AGEB85 years and older   SEXMale
WAISTNot at risk
Borderline      -0.7247957               -2.38838 1.0104620
-0.5139919
High            -3.5545436               -3.03945 0.8090797
-0.8383199
           WAISTSubstantially increased risk DSMKHHRBOne
DSMKHHRBTwo
Borderline                          0.4613057  -0.2101724
-0.5448591
High                                0.8062869  -0.1613995
-0.1612600
           DSMKHHRBThree or more
Borderline             0.5062501
High                  -0.0144048

Std. Errors:
           (Intercept) EXREGUICDid not meet recommended
guidelines
Borderline   0.5404158
0.09223007
High         0.4982189
0.08983821
           AGEB20-24 years AGEB25-29 years AGEB30-34 years
AGEB35-39 years
Borderline       0.6101939       0.5506572       0.5528414
0.5575772
High             0.5535413       0.5139154       0.5130832
0.5018881
           AGEB40-44 years AGEB45-49 years AGEB50-54 years
AGEB55-59 years
Borderline       0.5475781       0.5407680       0.5402762
0.5435554
High             0.5019232       0.4979563       0.4980880
0.4998473
           AGEB60-64 years AGEB65-69 years AGEB70-74 years
AGEB75-79 years
Borderline       0.5390645       0.5482030       0.5567464
0.6284221
High             0.4990328       0.5050975       0.5277893
```

```
0.5758979
          AGEB80–84 years AGEB85 years and older    SEXMale
WAISTNot at risk
Borderline       0.6308773               1.138947 0.09324778
0.1340679
High             1.1153564               1.119725 0.08977423
0.1487244
          WAISTSubstantially increased risk DSMKHHRBOne
DSMKHHRBTwo
Borderline                         0.1130690   0.1341583
0.2801941
High                               0.1124924   0.1253106
0.2249422
          DSMKHHRBThree or more
Borderline             0.6849079
High                   0.7988814

Residual Deviance: 6942.81
AIC: 7030.81
```

▼ Code

```
# GGT
model_file = here("Objects", "Full_data_MLR_GGT3.rds")
if (file.exists(model_file)) {
  full.mlr.ggt = readRDS(model_file)
} else {
  full.mlr.ggt = multinom(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST +
                          data = fdat.full, Hess = TRUE)
  saveRDS(full.mlr.ggt, model_file)
}
summary(full.mlr.ggt)
```

```
Call:
multinom(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST +
DSMKHHRB,
    data = fdat.full, Hess = TRUE)

Coefficients:
          (Intercept) EXREGUICDid not meet recommended
guidelines
Borderline  -15.348782
0.1641584
High         -4.161508
0.3609772
          AGEB20–24 years AGEB25–29 years AGEB30–34 years
AGEB35–39 years
```

```
Borderline      12.2297326       12.281339        12.17784
12.230075
High             0.6053798        1.046025         1.52910
1.478558
          AGEB40–44 years AGEB45–49 years AGEB50–54 years
AGEB55–59 years
Borderline       12.250944       12.435542       12.516705
12.436783
High              1.768828        1.896048        2.239024
2.041926
          AGEB60–64 years AGEB65–69 years AGEB70–74 years
AGEB75–79 years
Borderline       12.552362       12.279577       12.339176
12.169890
High              2.264161        2.222337        1.938956
2.036739
          AGEB80–84 years AGEB85 years and older     SEXMale
WAISTNot at risk
Borderline       12.524136              11.877897  0.32117474
-0.7101271
High              1.803366               2.100943 -0.02327147
-0.5035367
          WAISTSubstantially increased risk DSMKHHRBOne
DSMKHHRBTwo
Borderline                          0.2993336   0.5079295
0.4841456
High                                0.5552730   0.2116053
0.6224954
          DSMKHHRBThree or more
Borderline           -11.2477133
High                   0.8642364

Std. Errors:
          (Intercept) EXREGUICDid not meet recommended
guidelines
Borderline   0.1520737
0.12326681
High         1.0195482
0.08556511
          AGEB20–24 years AGEB25–29 years AGEB30–34 years
AGEB35–39 years
Borderline        0.373064       0.2529111       0.2514521
0.2225162
High              1.110124       1.0429165       1.0308648
1.0281012
          AGEB40–44 years AGEB45–49 years AGEB50–54 years
AGEB55–59 years
Borderline       0.2064138       0.1843157       0.1820639
```

```
0.1822491
High             1.0232944       1.0213494       1.0195544
1.0204282
           AGEB60–64 years AGEB65–69 years AGEB70–74 years
AGEB75–79 years
Borderline       0.1677742       0.2029949       0.2305137
0.2882291
High             1.0189988       1.0205605       1.0259927
1.0297450
           AGEB80–84 years AGEB85 years and older    SEXMale
WAISTNot at risk
Borderline       0.3043434              0.5585126 0.12144913
0.1907271
High             1.0414000              1.0574551 0.08415209
0.1375935
           WAISTSubstantially increased risk DSMKHHRBOne
DSMKHHRBTwo
Borderline                          0.1467620   0.1544587
0.2942175
High                                0.1062391   0.1163696
0.1993675
           DSMKHHRBThree or more
Borderline          5.625674e-07
High                5.990222e-01

Residual Deviance: 6114.484
AIC: 6202.484
```

I will leave the actual analysis for later; this shorter document is just meant to see what would happen if we did use the entire data set and whether it is really worth doing or not.

Next Step

I think the next step for our analysis would likely be to use the full adult data set, but use an indicator H for being in the sub-population of interest, and then use logistic regression with an interaction model (`EXREGUIC * H`) in order to measure e.g., how much the physical-activity effect changes when we move from the broader population into the smoker + medium/high-alcohol subgroup.